MUHAMMAD TAIMOOR ASHFAQ MALIK

GIS & Remote Sensing Specialist | GeoAI & Spatial Solutions Developer

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PROFESSIONAL SUMMARY

GIS and Remote Sensing professional specializing in spatial analysis, Python automation, and GeoAI solutions. Experienced in disaster risk assessment, climate applications, and geospatial solutions development using ArcGIS Pro, QGIS, PostGIS, C#, and Python. Passionate about building scalable geospatial tools that improve analytical workflows and decision-making.

WORK EXPERIENCE

Junior GIS Analyst | Nook Office Solutions · Islamabad, Pakistan *Oct 2025 – Present*

- Translated intricate urban planning codes into structured spatial data to support large-scale zoning analysis and mapping projects.
- Systematically integrated qualitative policy data into GIS datasets.
- Engineered custom Python-based tools and scripts to automate data cleaning, batch file creation, and cloud upload processes, significantly reducing manual processing time.
- Contributed to the iterative refinement of internal QA tools, improving the accuracy and validation speed of geospatial deliverables.

GIS Intern | National Disaster Risk Management Fund · Islamabad, Pakistan *Jun 2025 – Sep 2025*

- Digitized and managed geospatial datasets on Pakistan's energy infrastructure to support national disaster risk assessments.
- Conducted hazard assessments for multiple disaster scenarios to evaluate infrastructure vulnerability.
- Conducted flood extent assessment of the recent floods in Punjab to evaluate the magnitude and rarity of the event.
- Developed Python-based scripts for automated acquisition of satellite data and integrated outputs into a PostgreSQL/PostGIS database.
- Received hands-on training in React.js for developing interactive geospatial web applications.

Research Intern | Global Climate-Change Impact Studies Centre · Islamabad, Pakistan *Jun 2023 – Jul 2024*

- Conducted in-depth research on the impacts of climate change on various ecosystems and communities.
- Focused on the development of a RUSLE-based tool for analyzing climate-induced soil erosion and reservoir sedimentation in the Tarbela Region.
- Worked on mangrove restoration and suitability analysis using multiple MCDM techniques.
- Developed Python-based toolboxes to streamline analyses and geoprocessing tasks.

Cadastral Mapping Specialist | Gravity Solutions · Rawalpindi, Pakistan *Jun 2024 – Aug 2024*

- Digitized different types of land parcels.
- Ensured accuracy by performing quality checks and removing topological errors.
- Developed and maintained databases.
- Developed Python-based scripts to streamline processes.

PROJECTS

ArcGIS Pro Add-In for AI Ship Detection

- Engineered a custom ArcGIS Pro Add-In to automate maritime object detection across Synthetic Aperture Radar (SAR) and optical satellite imagery.
- Integrated a custom-trained YOLOv5 neural network into the ArcGIS geoprocessing pipeline, writing custom Python wrappers for SAR data normalization and bounding-box translation, and automated an end-to-end workflow chaining inference, Non-Maximum Suppression (NMS), coordinate projection, and geodesic area filtering.

AHP/FAHP Spatial Analyzer Add-in for ArcGIS Pro

- Developed a professional C#/.NET 8.0 Add-in for spatial MCDA featuring real-time Consistency Ratio (CR) validation, ArcPy integration for raster overlays, and a Monte Carlo Sensitivity Analysis module.

Aqua-Assist: GeoAI-Driven Spatial Decision Support System for Rainwater Harvesting

- Architected an agentic SDSS with a dual-pipeline framework to evaluate rainwater harvesting suitability, extracting hydrological parameters directly from raw DEMs and multispectral rasters.
- Engineered a Python-based FAHP integrated with an offline LLM (Ollama/LangChain) and Streamlit/Folium GUI to autonomously adjust spatial weights and deliver explainable GeoAI justifications.

National Spatial Data Infrastructure (NSDI) for Pakistan: Road Network Digitization

- Developed a refined road network dataset as a foundational component of Pakistan's NSDI, addressing fragmentation, incomplete metadata, and misaligned geometries.
- Conducted topology checks and quality assurance, producing a streamlined, navigable dataset supporting geospatial analysis and decision-making for sustainable development.

Disaster Management Plan for the Shishper Glacier GLOF, Pakistan

- Analyzed the 2022 Shishper Glacier outburst event using Landsat-8/9, Sentinel-2, MODIS, and ALOS PALSAR data to map thermal and debris changes, identifying rapid glacier melt triggers.
- Quantified post-flood infrastructure damage using satellite-derived indices to formulate a disaster management framework integrating early warning systems and climate-resilient mitigation strategies.

Digitization and Hazard Assessment of Energy Sector Data (NGCP)

- Digitized 27 transmission lines and 35 grid stations across Pakistan using QGIS and ArcGIS Pro, exporting shapefiles, KML, and a geodatabase.
- Overlaid assets with flood, cyclone, and earthquake hazard layers (50- and 495-year return periods) to score infrastructure vulnerability by severity.

Mapping Salt-Affected Soils Using Remote Sensing and GIS (Chakwal, Pakistan)

- Mapped soil salinity across Chakwal district using spectral indices (NDSI, NDVI, VSSI, SI1–SI4, Salinity Ratio) from Landsat 8/9 and Sentinel-2 imagery, classifying low, moderate, and high salinity zones.

Developing Shapefile to KML Converter Desktop Application

- Built a Python desktop app (tkinter, PyProj, SimpleKML) to load ZIP shapefiles, reproject to WGS84, and export KML files for Google Earth; packaged as a standalone executable with PyInstaller.

Identification of Potential Rainwater Harvesting Zones for Agricultural Production, Khyber Pakhtunkhwa

- Built a multi-criteria geospatial framework (ArcGIS, QGIS, Google Earth Engine, FAHP) to rank rainwater-harvesting sites by topography, soil, rainfall, and aridity, producing suitability hotspot maps for policymakers.

CORE SKILLS

GIS & Remote Sensing: ArcGIS Pro, QGIS, Erdas Imagine, ENVI, Trimble eCognition, Google Earth Engine, OSGeo4W, OSGeoLive, GDAL, AutoCAD Civil 3D, Spatial Decision Support Systems

GeoAI & Machine Learning: PyTorch (YOLOv5), Ollama & LangChain LLM Integration, Spatial MCDA / FAHP, Explainable AI Workflows, Agentic Spatial Decision Support Systems

Programming & Development: Python, C#, .NET 8 Development, ArcGIS Pro SDK, ArcPy, JavaScript, Node.js, React.js, Django Web Framework

Databases & Platforms: PostgreSQL/PostGIS, MySQL, ETL Operations

Web Visualization & Services: Leaflet.js, OpenLayers.js, d3.js, Streamlit, GeoServer, TiTiler, FastAPI, Mapbox GL, MapLibre, GeoLibre, GeoLens, Jupyter

DevOps & Tools: PyInstaller, Linux, Docker, Virtualization Platforms (VMware, Hyper-V)

EDUCATION

Master's in Remote Sensing & GIS *2025 – Present*
National University of Science & Technology, Islamabad

BS Geoinformatics *2021 – 2025*
PMAS Arid Agriculture University, Rawalpindi
Final Grade: 3.61 / 4.0 | Thesis: Developing a Web-Based Tool for Spatio-Temporal Crop Mapping using Sentinel Imagery in the Punjab Region

PUBLICATIONS

Developing a RUSLE-Based Tool for Analyzing Climate Induced Soil Erosion and Reservoir Sedimentation in the Tarbela Region (2026)

Muhammad Arif Goheer, Sher Shah Hassan, Muhammad Taimoor Ashfaq Malik, Syeda Rosheen Raza Kazmi, Syed Zee Ali Shan — International Journal of Environmental Research, Springer Nature

CERTIFICATIONS

- Introduction to GIS Mapping — University of Toronto | Coursera
- Fundamentals of GIS — UC Davis | Coursera
- Going Places With Spatial Analysis — ESRI MOOC
- Make An Impact with Modern GeoApps — ESRI MOOC
- Spatial Data Science: The New Frontier in Analytics — ESRI MOOC
- Basics of AutoCAD Civil 3D — Department of Land & Water Conservation Engineering, PMAS Arid Agriculture University Rawalpindi